Goal 1: An Infrequent Semantic Relation Learns a Shared Affine Representation Through a Nonlinear Lookup

factual-recall-mechanisms

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1 Result

We train a small transformer for one epoch on a synthetic corpus containing a semantic relation R_{sem} and a frequency-matched random control R_{rnd}. At $F{=}16$ occurrences per triple, R_{sem} reaches recall 0.672 on trained triples and 0.635 on subjects absent from all relation triples. An ordinary least-squares map from subject embeddings to object embeddings also selects the correct held-out object with accuracy 0.990 ± 0.007 across three seeds; chance is 0.03.

The representation contains the affine structure. For the fixed-relation map from subject residuals to object logits, the affine fit has $R^2 = .965 \pm .010$ and relative error $.185 \pm .027$. The same map has midpoint defect $.352 \pm .011$ and exact-Jacobian dispersion $.281 \pm .013$, both nonzero. The executed lookup is therefore nonlinear and not J-equivalent on the tested domain. Attribute training creates the shared affine geometry even at $F{=}1$, well before the transformer recalls the relation reliably.

2 Data

Vocabulary (265 tokens). A separator `x`; two relation tokens `r_0`, `r_1`; 192 subject tokens `a_0`, ..., `a_191`; 32 object tokens `b_0`, ..., `b_31`; 32 attribute tokens `f_0`, ..., `f_31`; 6 attribute tokens `g_0`, ..., `g_5`.

Entities and attributes. Subject `a_i` has attribute pair

$$(f_{i \bmod 32},\ g_{\lfloor i/32 \rfloor}).$$

The 192 subjects cover each of the 32×6 combinations once. Object `b_j` has attribute pair

$$(f_{(j+3) \bmod 32},\ g_0).$$

Sentences. All training sentences have five tokens:

attribute sentence	`x a_i f_p g_q x`	and	`x b_j f_p g_q x`
relation triple	`x a_i r_r b_t x`		

Relations.

$$R_{\text{sem}}(a_i) = b_{f(i)}, \qquad R_{\text{rnd}}(a_i) = b_{\rho(i)},$$

Here $f(i) = i \bmod 32$ is the subject's f-attribute index. The semantic target is the object whose shifted f-attribute matches that index. The function ρ is a seed-dependent balanced random assignment to the same 32 objects and is independent of the attributes. For example, in seed 19, `a_37` has attributes (`f_5`, `g_1`). Its semantic target is `b_5`, whose attribute sentence reads `x b_5 f_8 g_0 x`, while its random target is `b_27`. The corresponding relation sentences are `x a_37 r_0 b_5 x` and `x a_37 r_1 b_27 x`.

Split and exact frequency matching. For each `f_`-value, four of its six subjects enter relation training and two are held out. This gives 128 trained and 64 held-out subjects, with a seed-dependent split. The random assignment ρ is balanced within both groups. Under either relation, every trained subject occurs in one triple, and every object is the target of four trained subjects and two held-out subjects. Each relation contributes 128 training triples, repeated F times. The relations therefore have identical frequency statistics. Held-out subjects occur in attribute sentences but in no relation triple. The test `tests/test_linear_shift.py` checks these invariants.

3 Model and training

Model	causal transformer, 1 layer, hidden 64, 4 heads, context 8
Embeddings	input and output embeddings untied; all weights $\mathcal{N}(0, 0.02^2)$
	no attribute-based initialization; nothing frozen; no loss reweighting
Parameters	84,288
Corpus	every attribute sentence $\times A$=128 (= 28,672 sentences)
	+ 128 triples per relation, each $\times F$ ($F \in \{1, 4, 16, 64\}$)
Training	**exactly one epoch**: a single pass over the shuffled corpus
Steps	452 (F=1) to 704 (F=64) steps at batch size 64
Objective	unweighted next-token cross-entropy on all positions
Optimizer	AdamW, constant LR 10^{-3} after 5 linear warmup steps,
	weight decay 0, gradient clip 1.0
Seeds	19, 31, 47 (control data assignment, initialization, and data order)

At F=1 every relation triple occurs exactly once in the entire training history of the model.

4 Metrics

Shared-rule probe. The representation-level diagnostic measures whether one affine rule transfers to subjects excluded from relation training:

> Fit $\hat{W}, \hat{b}$ by ordinary least-squares linear regression from the subject's *input embedding* $s_i \in \mathbb{R}^{64}$ to the target object's *output embedding* (unembedding row) $o_{R(i)} \in \mathbb{R}^{64}$, using the 128 trained pairs. For each of the 64 held-out subjects, predict $\hat{W}s_j + \hat{b}$ and select the nearest object row by cosine similarity among all 32 objects. Report the fraction of held-out subjects for which the selected object is $R(j)$. Chance is $1/32 \approx 0.03$.

Each subject has one correct object under each relation, so top-k metrics are unnecessary. We separately record greedy next-token recall on prompts `x a_i r_r` for trained and held-out

subjects. Recall checks that the model learned the relation facts. The OLS probe instead tests whether the subject and object embedding spaces admit one affine decoder that transfers to unseen relation subjects. It measures representation geometry and generalization; it does not classify the transformer's computation as affine.

Transformation-level affine and J-equivalence tests. For a fixed relation token, each learned sublayer defines a continuous map on its residual-stream coordinates. We test six maps: normalized subject state to attention update; subject residual to post-attention residual; normalized post-attention state to MLP update; post-attention residual to post-MLP residual; post-MLP residual to centered object logits; and subject residual to centered object logits. These are the attention core, attention block, MLP core, MLP block, readout, and full lookup.

For each map, we report a four-fold target-stratified affine reconstruction on the 128 trained subjects, normalized midpoint defect, and dispersion of exact Jacobians across subjects. An affine map has zero midpoint defect and a constant Jacobian. We also report decision accuracy after replacing the map output with its affine prediction, but this substitution does not prove affinity. On a synthetic affine control, the implementation gives midpoint defect below 10^{-6} and Jacobian dispersion below 10^{-12}.

5 Results

		probe accuracy (held-out)		recall (trained)		recall (held-out)	
	F	R_{sem}	R_{rnd}	R_{sem}	R_{rnd}	R_{sem}	R_{rnd}
	1	$1.000 \pm .000$	$.031 \pm .013$	.029	.010	.016	.010
	4	$1.000 \pm .000$	$.016 \pm .013$	.055	.034	.057	.031
	16	$\mathbf{0.990 \pm .007}$	$.036 \pm .007$	**.672**	.167	**.635**	.031
control: no triples		$1.000 \pm .000$	$.021 \pm .015$	–	–	.000	.000

Table 1: Low-frequency results, reported as mean ± sd over seeds 19/31/47 at A=128. The value F is the number of occurrences of each of the 128 triples per relation. At F=16, R_{sem} reaches recall .672 on trained subjects and .635 on held-out subjects, while its affine embedding decoder transfers at .990. The frequency-matched R_{rnd} is barely recalled and shows no shared-rule transfer. The table concerns learning and representation geometry, not the affinity of the executed transformation.

Low-frequency relation learning (F=16). With each fact occurring 16 times in one epoch, R_{sem} reaches recall .672 on trained triples and .635 on held-out subjects. The embedding probe transfers at 0.990. The frequency-matched random control reaches only .167 recall and its probe stays at chance. Shared attribute geometry lets the semantic relation learn and generalize earlier. These measurements concern structure and transfer.

Affine geometry after one occurrence (F=1). At F=1, the probe is already 1.000 for R_{sem} and remains at chance for R_{rnd}. The model still fails to recall individual triples at this point ($\approx .03$), so this is a representation result rather than a behavioral one.

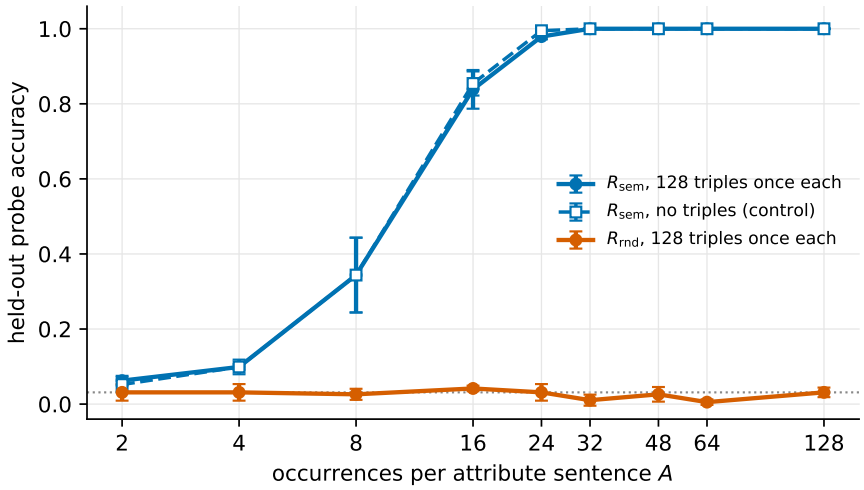


Figure 1: Held-out shared-rule probe accuracy at $F{=}1$ as attribute-sentence frequency A increases. Every relation triple occurs once in the solid curve; the dashed control contains no relation triples. Both curves undergo the same transition from chance to affine decodability. The random relation has no transferable counterpart.

Attribute training drives the decodability transition. At $F{=}1$, increasing the attribute-sentence frequency changes the R_{sem} probe from 0.05 to 0.10, 0.34, 0.85, and 1.00 for $A = 2, 4, 8, 16, 24$ (Fig. 1). The zero-triple control follows the same curve. Attribute sentences therefore build the affine subject-to-object geometry. The single-occurrence semantic triples agree with that geometry, and the model uses it once recall emerges.

Map	affine R^2	rel. error	affine decision	midpoint defect	Jacobian disp.
attention core	$.993 \pm .002$	$.080 \pm .012$	$.742 \pm .072$	$.035 \pm .007$	$.085 \pm .013$
attention block	$.983 \pm .004$	$.129 \pm .015$	$.750 \pm .066$	$.273 \pm .010$	$.179 \pm .005$
MLP core	$.983 \pm .003$	$.131 \pm .011$	$.742 \pm .061$	$.095 \pm .005$	$.137 \pm .007$
MLP block	$.974 \pm .006$	$.160 \pm .017$	$.758 \pm .057$	$.247 \pm .027$	$.172 \pm .009$
readout	$.993 \pm .003$	$.081 \pm .017$	$.760 \pm .060$	$.199 \pm .004$	$.228 \pm .007$
full lookup	$.965 \pm .010$	$.185 \pm .027$	$.750 \pm .056$	$.352 \pm .011$	$.281 \pm .013$

Table 2: Transformation tests for R_{sem} at $F{=}16$. Affine fits use four target-stratified folds, so every trained subject is evaluated once. An affine or J-equivalent map would have zero midpoint defect and zero Jacobian dispersion. All six maps are nonlinear by these tests, although the quality of their affine approximations differs.

The semantic lookup is nonlinear. The full lookup has midpoint defect $.352 \pm .011$ and exact-Jacobian dispersion $.281 \pm .013$. Dispersion remains $.284 \pm .013$ on the empirical subject-difference subspace, ruling out an explanation based only on off-manifold directions. The high affine R^2 and transferred embedding decoder show a close affine approximation. The nonzero defects show that the trained transformer is neither affine nor J-equivalent on the tested domain.

6 Learned representation and causal mechanism

Attribute training organizes the subject embeddings into 32 code families, one for each $f(i)$ value, instead of storing the 128 semantic triples independently. A target-class centroid learned from the 128 trained subjects classifies the embeddings of the 64 held-out relation subjects with accuracy $1.000 \pm .000$. The class remains decodable from concatenated attention-head outputs at $1.000 \pm .000$, from the post-attention state at $.990 \pm .007$, and from the final state at $.990 \pm .007$. The model represents $f(i)$ as a distributed direction in the subject embedding. Subjects with the same f value share that task-relevant component while retaining identity and g information.

Each object has a learned output-embedding direction. The relation-conditioned MLP and output readout convert the transported f-class code to the corresponding object direction. The pseudocode below records the causal information flow. It does not assume that the component maps are affine.

```
def answer_semantic(subject):
    # Learned representation: a shared code for f(subject).
    subject_code = embedding[subject]

    # H3 and H1 carry most of the subject's value payload;
    # H0/H2 provide smaller, seed-dependent support.
    payload = sum(head[h].read_value(subject_code, relation=r0)
                  for h in [H0, H1, H2, H3])

    # r0 selects the shared class-to-object readout.
    state = relation_state[r0] + project(payload)
    state += distributed_class_readout(state)
    object_scores = dot(final_normalize(state), object_directions)
    return argmax(object_scores)          # b[f(subject)]
```

In the pseudocode, `read_value` multiplies the subject's projected value by the attention weight assigned to that subject. Separator and relation contributions do not change under a subject swap and belong to the baseline. The final `dot` compares the resulting state with 32 learned object-output directions.

The interventions locate the information in the value payload. Swapping that payload between two subjects restores $.987 \pm .008$ of the corresponding change in the correct-object logit; swapping only their attention weights restores $.013 \pm .007$. A swap of all head outputs restores the full effect. Replacing every head output with its training-subject mean reduces object accuracy from $.729 \pm .094$ to chance ($.031 \pm .000$). The downstream computation therefore uses the class code detected by the probes.

The relation token selects how the model interprets the transported subject code. When the subject is fixed and `r_0` is exchanged with `r_1`, a head-output patch restores $.070 \pm .007$ of the relation effect, while an MLP-population patch restores $.850 \pm .017$. The heads primarily move subject content; the `r_0` residual selects the semantic readout downstream.

The final conversion is distributed across the MLP. Keeping only the 32 units with the largest causal effects recovers $.498 \pm .029$ of the subject-specific margin. Ablating those units still leaves $.790 \pm .032$, so no small group carries the computation alone. An affine surrogate for every GELU preserves held-out object accuracy ($.729 \pm .092$ versus $.729 \pm .094$) and explains $.962 \pm .001$ of activation variance. This intervention provides an accurate approximation and preserves the decisions, but it does not reproduce the MLP transformation exactly. The MLP core has midpoint defect

.095 ± .005 and Jacobian dispersion .137 ± .007; the full MLP block gives .247 ± .027 and .172 ± .009. Both maps are nonlinear under the affine/Jacobian definition.

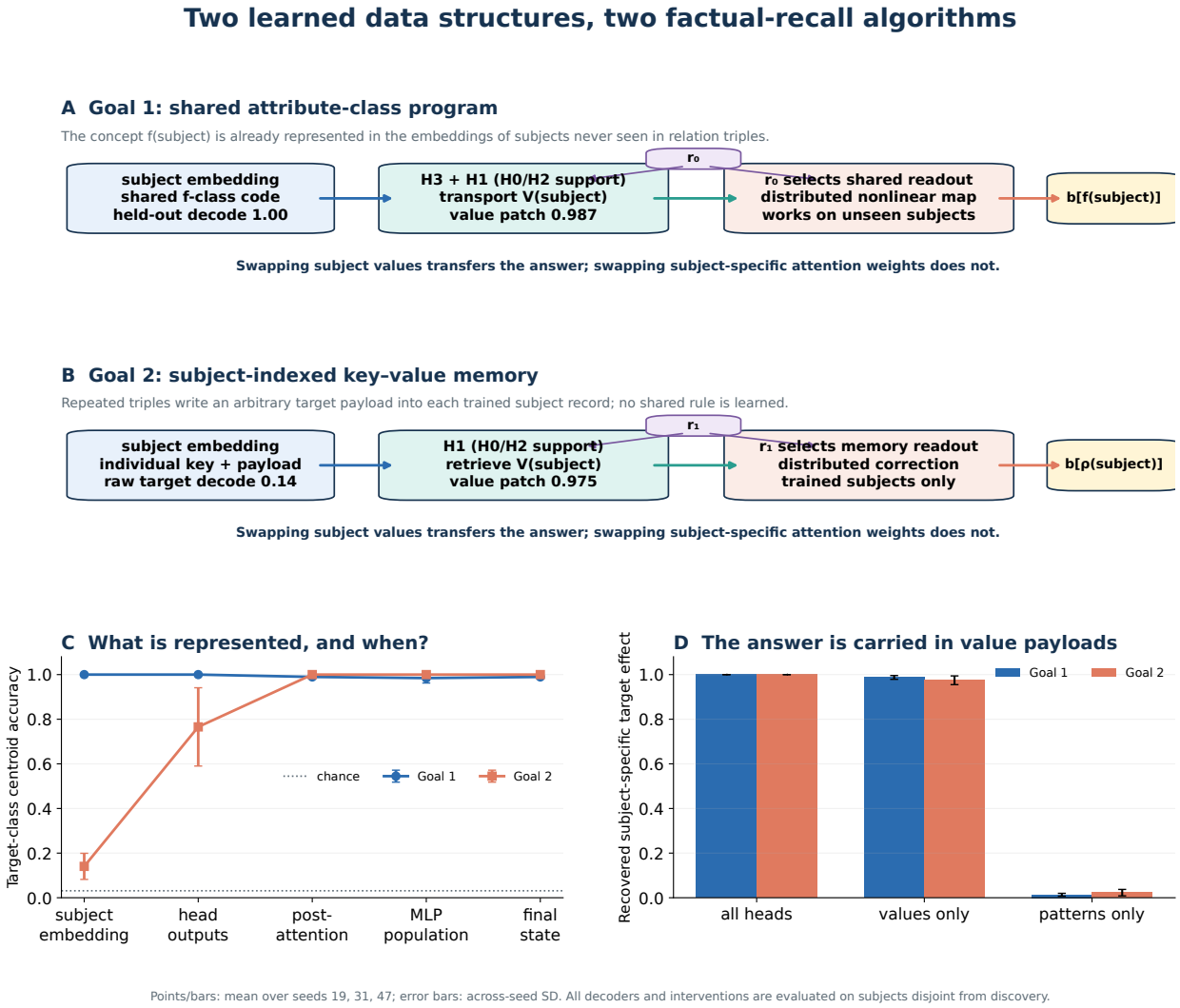


Figure 2: Learned representations and causal information flow in the original transformer experiments. For Goal 1, the model transports a shared f-class code to a distributed class-to-object readout. The lower panels track target decodability and compare the causal effects of attention values and attention weights.

The probes and interventions support one account of generalization. Held-out subject embeddings contain the shared f-organized code; the attention value paths transport it; and a distributed nonlinear MLP/readout converts it to the correct object direction. On these prompts, an affine model closely approximates that computation without matching it exactly.

7 Scope

At F=16, the semantic relation generalizes through a shared, affine-decodable representation. Its attention, MLP, readout, and full lookup are neither affine nor J-equivalent on the tested domain.

The semantic lookup is more closely approximated by an affine map than the frequency-matched random lookup, and affine GELU substitution preserves its decisions. Approximation, decision equivalence, and exact affinity remain separate properties.

The relation triples do not cause the decodability transition at $F{=}1$. The dashed control in Fig. 1 attributes that transition to exposure to attribute sentences. We also tested three designs in which transfer could only come from triple-driven structure: a one-to-one relation with separate output rows for each fact, a conjunction of two attributes with up to 49,152 distinct facts, and hidden-state probes. None transferred above chance in the single-occurrence, one-epoch regime, where each fact contributes one gradient.

Earlier drafts used affine decoding and generalization as the definition of a linear relation. Here those measurements support a claim about shared representation; midpoint and exact-Jacobian tests classify the computation. The reported "single-exposure phase shift" in those drafts, from 0.27 to 0.95, also disappears under a fixed-size probe and standard initialization. It arose because the earlier regression used 10 pairs at the first snapshot and more than 500 later, over embeddings initialized from the attribute geometry.